Function

```
load_case = 2;
design = 2;
function [V,m_max, FOS_1, FOS_2, FOS_3, FOS_4, FOS_5, FOS_6, FOS_7, FOS_8]
= failure_specs(b,h,y_i,load_case,P,pos_case, design, section)
    s_ten = 30;
    s_comp = 6;
    s_shear = 4;
    s_glue = 2;
    E = 4000;
    v = 0.2;

    height = y_i(end) + h(end);

    %Setting width between webs
    wid = 80;

    %Calculate cross-sectional properties
    y = y_i + h/2;
    A_total = dot(b,h);
    ybar = dot(b.*h, y)/A_total;
    ybar_top = height - ybar;
    d = abs(ybar-y);
    I = dot(b,h.^3)/12 + dot(b.*h,d.^2);

    k_1 = 4;
    k_2 = 0.425;
    k_3 = 6;
    k_4 = 5;

    %Set max moments, safety
    if load_case == 2
        if pos_case == 1 %Section C -> 285 - 915 mm
            pos = 110.2;
            m_max = 78099;
        elseif pos_case == 2 %Section B -> 100 - 285 mm & 915 - 1130 mm
            pos = 203;
            m_max = 40800;
        elseif pos_case == 3 %Section A -> 0 - 100 mm & 1100 - 1200
            pos = 95;
            m_max = 23000;
        end
    elseif load_case == 1
        pos = 128;
        m_max = 69400;
    elseif load_case == 3
        pos = 116.7;
        m_max = 80900;
    end
```

```
    fbd = zeros(8, 2);
    fbd(:,2) = [0, pos, pos+176. pos+340, pos+340+176, pos+340+176+164,
pos+340+176+164+176, 1200]';
    if load_case == 1
        P = 400;
        fbd(2:7,1) = -P/6;
    else
        fbd(:,1) = [0, -67.5, -67.5, -67.5, -67.5, -91, -91, 0];

    end


    % Reaction Forces
    Hy = dot(fbd(:,1), fbd(:,2))/-1200;
    Ay = -sum(fbd(:,1))-Hy;

    %Flexural Compression FOS
    FOS_1 = s_comp*I/(ybar_top*m_max);

    %Flexural Tension FOS
    FOS_2 = (s_ten*I)/(ybar*m_max);

    %Maximum Shear
    if load_case == 1
        V = 257;
    else
        V = 268;
    end

    %find sections which the y_bar crosses
    rows_bar = ybar > y_i & ybar < h+y_i;
    w = sum(b(rows_bar));
    Q = sum(b(rows_bar).*(ybar^2)/2) + b(1)*h(1)*(ybar-(h(1)/2));
    FOS_3 = (s_shear*I*w)/(Q*V);

    %Glue shear FOS
    Q_glue = b(end)*h(end)*(ybar_top-h(end)/2);
    w = 2*b(end-1);
    FOS_4 = (s_glue*I*w)/(Q_glue*V);

    %Case 1 Plate Buckling
    b_1 = wid + 1.27;
    if section == "A"
        t_1 = 1.27;
    elseif section == "B"
        t_1 = 2.54;
    elseif section == "C"
        t_1 = 3.81;
    else
```

```
        t_1 = h(end);
    end

    stress_1 = ((k_1*pi*pi*E)/(12*(1-v^2))) * (t_1/b_1)^2;
    FOS_5 = (stress_1*I)/(m_max*ybar_top);

    %splitting top into multiple sections
    b_2 = (b(end)-b_1)/2;
    t_2 = h(end);

    %Case 2 Plate Buckling
    stress_2 = ((k_2*pi*pi*E)/(12*(1-v^2))) * (t_2/b_2)^2;
    FOS_6 = (stress_2*I)/(m_max*ybar_top);

    %Case 3 Plate Buckling
    t_3 = b(rows_bar);
    b_3 = (h + y_i - ybar);
    b_3 = b_3(rows_bar);
    stress_3 = ((k_3*pi*pi*E)/(12*(1-v^2))) * (t_3(1)/b_3(1))^2;
    FOS_7 = (stress_3*I)/(m_max*ybar_top);

    %Case 4 Plate Buckling
    a = 200;
    height = h(rows_bar);
    height = height(1);
    t_4 = 1.27;
    b_4 = sum(b(rows_bar));
    stress_4 = ((k_4*pi*pi*E)/(12*(1-v^2))) * ((t_4/a)^2 + (t_4/height)^2);
    FOS_8 = (stress_4*I*b_4)/(Q*V);
end
```

Graphing

```
%P = 400;

%Change Load Case


points = 50;

train = linspace(-856,1200,points);
x_values = linspace(0,1200,points);
for i = 1:length(train)
    pos = train(i);

    %Free Body Diagram - with forces and distance
    fbd = zeros(8, 2);
    if load_case == 2
        fbd(:,1) = [0, -67.5, -67.5, -67.5, -67.5, -91, -91, 0];
```

```
    elseif load_case == 1
        fbd(2:7,1) = -400/6;
    elseif load_case == 3
        fbd(:,1) = [0, -67.5*1.1, -67.5*1.1, -67.5, -67.5, -91, -91, 0];
    end

    %Set position of all wheels based on the wheel on the far left of the
    %train
    fbd(:,2) = [0, pos, pos+176. pos+340, pos+340+176, pos+340+176+164,
pos+340+176+164+176, 1200]';

    %Negate any wheels outside the bridge
    outer_rows = fbd(:,2) < 0 | fbd(:,2) > 1200;
    fbd(outer_rows, :) = [];

    %Reaction Forces
    Hy = dot(fbd(:,1), fbd(:,2))/-1200;
    Ay = -sum(fbd(:,1))-Hy;

    %Update Free-Body Diagram
    fbd(end, 1) = Hy;
    fbd(1,1) = Ay;

    ColOrd = get(gca,'ColorOrder');
    [m,n] = size(ColOrd);
    ColRow = rem(i,m);
        if ColRow == 0
          ColRow = m;
        end

    hold on

    %Generate Values for Moment and Shear
    sfd = cumsum(fbd(:, 1));

    bmd = [0; cumsum(sfd(1:end-1).*(fbd(2:end,2)-fbd(1:end-1, 2)))];


    %Graph BMD
    [m, idx] = max(bmd);
    hold on
    subplot(2,3,1)
    xlabel("Length along Bridge (mm)")
    ylabel("Bending Moment (Nmm)")
    plot(fbd(:,2),-bmd, "-");
    hold off
    hold on
    subplot(2,3,2)
    xlabel("Length along Bridge (mm)")
    ylabel("Bending Moment (Nmm)")
```

```
    plot(fbd(:,2),-bmd, "-");
    hold off
    hold on
    subplot(2,3,3)
    xlabel("Length along Bridge (mm)")
    ylabel("Bending Moment (Nmm)")
    plot(fbd(:,2),-bmd, "-");
    hold off

    %Graph SFD
    hold on
    subplot(2,3,4)
    xlabel("Length along Bridge (mm)")
    ylabel("Shear Force (N)")
    for n = 1:length(fbd(:,1))-1
        Col = ColOrd(ColRow,:);
        plot(fbd(n:n+1, 2), ones(2,1)*sfd(n), "Color", Col)
        hold on
        tiny = [fbd(n+1,2);fbd(n+1,2)+0.00001];
        plot(tiny, ones(2,1).*sfd(n:n+1), "Color", Col)
    end
    hold off
    hold on
    subplot(2,3,5)
    xlabel("Length along Bridge (mm)")
    ylabel("Shear Force (N)")
    for n = 1:length(fbd(:,1))-1
        Col = ColOrd(ColRow,:);
        plot(fbd(n:n+1, 2), ones(2,1)*sfd(n), "Color", Col)
        hold on
        tiny = [fbd(n+1,2);fbd(n+1,2)+0.00001];
        plot(tiny, ones(2,1).*sfd(n:n+1), "Color", Col)
    end
    hold off
    hold on
    subplot(2,3,6)
    xlabel("Length along Bridge (mm)")
    ylabel("Shear Force (N)")
    for n = 1:length(fbd(:,1))-1
        Col = ColOrd(ColRow,:);
        plot(fbd(n:n+1, 2), ones(2,1)*sfd(n), "Color", Col)
        hold on
        tiny = [fbd(n+1,2);fbd(n+1,2)+0.00001];
        plot(tiny, ones(2,1).*sfd(n:n+1), "Color", Col)
    end
    hold off
end
```

Function Calling + FOS Graphing

```
pos_case = 1;
if design == 0
    %Design 0 Cross-Sectional Dimensions
    b = [77.46;1.27;1.27;5;5;100];
    h = [1.27;75;75;1.27;1.27;1.27];
    y_i = [0;0;0;73.73;73.73;75];
    [V, m_max, FOS_1, FOS_2, FOS_3, FOS_4, FOS_5, FOS_6, FOS_7, FOS_8] =
failure_specs(b,h,y_i,load_case,P,pos_case, design, "0");
else
    %Final Design Cross-Sectional Dimensions - Section B
    b = [38.73;1.27;1.27;5;5;80;5;5;120];
    h = [1.27;98.73;98.73;1.27;1.27;1.27;1.27;1.27;1.27];
    y_i = [0;0;0;1.27;1.27;97.46;97.46;97.46;98.73];
    [V, m_maxB,FOS_1B, FOS_2B, FOS_3B, FOS_4B, FOS_5B, FOS_6B, FOS_7B,
FOS_8B] = failure_specs(b,h,y_i,load_case,P,1, design, "B");
    ratio_B = 78099/40800;

    %Final Design Cross-Sectional Dimensions - Section C
    b = [38.73;1.27;1.27;5;5;80;70;5;5;120];
    h = [1.27;98.73;98.73;1.27;1.27;1.27;1.27;1.27;1.27;1.27];
    y_i = [0;0;0;1.27;1.27;96.19;97.46;97.46;97.46;98.73];
    [V, m_maxC,FOS_1C, FOS_2C, FOS_3C, FOS_4C, FOS_5C, FOS_6C, FOS_7C,
FOS_8C] = failure_specs(b,h,y_i,load_case,P,1, design, "C");

    %Final Design Cross-Sectional Dimensions - Section A
    b = [38.73;1.27;1.27;5;5;5;5;120];
    h = [1.27;98.73;98.73;1.27;1.27;1.27;1.27;1.27];
    y_i = [0;0;0;1.27;1.27;97.46;97.46;98.73];
    [V,m_maxA, FOS_1A, FOS_2A, FOS_3A, FOS_4A, FOS_5A, FOS_6A, FOS_7A,
FOS_8A] = failure_specs(b,h,y_i,load_case,P,1, design, "A");
    ratio_A = 78099/23000;

    disp('SECTION A:')
    tension_FOS = FOS_2A*ratio_A
    compression_FOS = FOS_1A*ratio_A
    shear_FOS = FOS_3A
    glue_FOS = FOS_4A
    buckling1_FOS = FOS_5A*ratio_A
    buckling2_FOS = FOS_6A*ratio_A
    buckling3_FOS = FOS_7A*ratio_A
    buckling4_FOS = FOS_8A
    disp('SECTION B:')
    tension_FOS = FOS_2B*ratio_B
    compression_FOS = FOS_1B*ratio_B
    shear_FOS = FOS_3B
    glue_FOS = FOS_4B
    buckling1_FOS = FOS_5B*ratio_B
    buckling2_FOS = FOS_6B*ratio_B
    buckling3_FOS = FOS_7B*ratio_B
    buckling4_FOS = FOS_8B
```

```
    disp('SECTION C:')
    tension_FOS = FOS_2C
    compression_FOS = FOS_1C
    shear_FOS = FOS_3C
    glue_FOS = FOS_4C
    buckling1_FOS = FOS_5C
    buckling2_FOS = FOS_6C
    buckling3_FOS = FOS_7C
    buckling4_FOS = FOS_8C


    if pos_case == 2
        secA_locations = [0 100 ; 1100 1200]'
        secB_location = [100 1100]'
    elseif pos_case == 3
        secA_locations = [0 400 ; 800 1200]'
        secB_location = [400 800]'
    end

end
```

```
SECTION A:
tension_FOS =
15.1953
compression_FOS =
4.6220
shear_FOS =
3.5250
glue_FOS =
8.8172
buckling1_FOS =
2.5787
buckling2_FOS =
4.8256
buckling3_FOS =
17.3266
buckling4_FOS =
3.1074
SECTION B:
tension_FOS =
9.0310
compression_FOS =
3.6595
shear_FOS =
3.4813
glue_FOS =
12.4241
buckling1_FOS =
8.1666
buckling2_FOS =
3.8207
buckling3_FOS =
20.0322
buckling4_FOS =
3.0688
SECTION C:
tension_FOS =
4.8231
compression_FOS =
```

```
2.3506
shear_FOS =
3.4296
glue_FOS =
15.3169
buckling1_FOS =
11.8029
buckling2_FOS =
2.4542
buckling3_FOS =
16.7795
buckling4_FOS =
3.0233
```

```
hold on

%Plot calculated FOS values onto the envelope
if design == 0
    subplot(2,3,1)
    plot([0,1200],[-FOS_1,-FOS_1]*m_max, "r-")
    plot([0,1200],[-FOS_2,-FOS_2]*m_max, "b-")
    subplot(2,3,2)
    plot([0,1200],[-FOS_5,-FOS_5]*m_max, "m-")
    plot([0,1200],[-FOS_6,-FOS_6]*m_max, "g-")
    subplot(2,3,3)
    plot([0,1200],[-FOS_7,-FOS_7]*m_max, "c-")

    subplot(2,3,4)
    plot([0,1200],[-FOS_3,-FOS_3]*257, "r")
    plot([0,1200],[FOS_3,FOS_3]*257, "r")

    subplot(2,3,5)
    plot([0,1200],[-FOS_4,-FOS_4]*257, "c")
    plot([0,1200],[FOS_4,FOS_4]*257, "c")

    subplot(2,3,6)
    plot([0,1200],[-FOS_8,-FOS_8]*257, "m")
    plot([0,1200],[FOS_8,FOS_8]*257, "m")


else

    subplot(2,3,1)
    comp = plot(secA_locations(:,1),[-FOS_1A,-FOS_1A]*m_maxA, "r-");
    plot([100,100.0001], [-FOS_1A, -FOS_1B]*m_maxA, "r")
    plot([100, 285],[-FOS_1B,-FOS_1B]*m_maxB, "r-")
    plot([285, 285],[-FOS_1B,-FOS_1C]*m_maxC, "r-")
    plot([285,915], [-FOS_1C,-FOS_1C]*m_maxC, "r-")
    plot([915, 915],[-FOS_1B,-FOS_1C]*m_maxB, "r-")
    plot([915,1100],[-FOS_1B,-FOS_1B]*m_maxB, "r-")
    plot([1100,1100.0001], [-FOS_1A, -FOS_1B]*m_maxA, "r")
    plot(secA_locations(:,2),[-FOS_1A,-FOS_1A]*m_maxA, "r-")
```

```
    tens = plot(secA_locations(:,1),[-FOS_2A,-FOS_2A]*m_maxA, "b-");
    plot([100,100.0001], [-FOS_2A, -FOS_2B]*m_maxB, "b")
    plot([100, 285],[-FOS_2B,-FOS_2B]*m_maxB, "b-")
    plot([285,285],[-FOS_2B,-FOS_2C]*m_maxC, "b-")
    plot([285,915], [-FOS_2C,-FOS_2C]*m_maxC, "b-")
    plot([915,915],[-FOS_2B,-FOS_2C]*m_maxC, "b-")
    plot([915,1100],[-FOS_2B,-FOS_2B]*m_maxB, "b-")
    plot([1100,1100.0001], [-FOS_2A, -FOS_2B]*m_maxB, "b")
    plot(secA_locations(:,2),[-FOS_2A,-FOS_2A]*m_maxA, "b-")

    legend([comp tens],{'Matboard Compression Failure','Matboard Tension
Failure'})


    subplot(2,3,2)
    case1 = plot(secA_locations(:,1),[-FOS_5A,-FOS_5A]*m_maxA, "m-");
    plot([100,100.0001], [-FOS_5A, -FOS_5B]*m_maxA, "m")
    plot([100,285],[-FOS_5B,-FOS_5B]*m_maxB, "m-")
    plot([285,285],[-FOS_5B,-FOS_5C]*m_maxB, "m-")
    plot([285,915], [-FOS_5C,-FOS_5C]*m_maxB, "m-")
    plot([915,915],[-FOS_5B,-FOS_5C]*m_maxB, "m-")
    plot([915,1100],[-FOS_5B,-FOS_5B]*m_maxB, "m-")
    plot([1100,1100.0001], [-FOS_5A, -FOS_5B]*m_maxA, "m")
    plot(secA_locations(:,2),[-FOS_5A,-FOS_5A]*m_maxA, "m-")

    case2 = plot(secA_locations(:,1),[-FOS_6A,-FOS_6A]*m_maxA, "g-");
    plot([100,100.0001], [-FOS_6A, -FOS_6B]*m_maxA, "g")
    plot([100,285],[-FOS_6B,-FOS_6B]*m_maxB, "g-")
    plot([285,285],[-FOS_6B,-FOS_6C]*m_maxB, "g-")
    plot([285,915], [-FOS_6C,-FOS_6C]*m_maxB, "g-")
    plot([915,915],[-FOS_6B,-FOS_6C]*m_maxB, "g-")
    plot([915,1100],[-FOS_6B,-FOS_6B]*m_maxB, "g-")
    plot([1100,1100.0001], [-FOS_6A, -FOS_6B]*m_maxA, "g")
    plot(secA_locations(:,2),[-FOS_6A,-FOS_6A]*m_maxA, "g-")

    legend([case1 case2],{'Case 1 Plate Buckling Failure','Case 2 Plate
Buckling Failure'})

    subplot(2,3,3)
    case3 = plot(secA_locations(:,1),[-FOS_7A,-FOS_7A]*m_maxA, "c-");
    plot([100,100.0001], [-FOS_7A, -FOS_7B]*m_maxA, "c")
    plot([100,285],[-FOS_7B,-FOS_7B]*m_maxB, "c-")
    plot([285,285],[-FOS_7B,-FOS_7C]*m_maxB, "c-")
    plot([285,915], [-FOS_7C,-FOS_7C]*m_maxB, "c-")
    plot([915,915],[-FOS_7B,-FOS_7C]*m_maxB, "c-")
    plot([915,1100],[-FOS_7B,-FOS_7B]*m_maxB, "c-")
    plot([1100,1100.0001], [-FOS_7A, -FOS_7B]*m_maxA, "c")
    plot(secA_locations(:,2),[-FOS_7A,-FOS_7A]*m_maxA, "c-")
```

```
legend([case3],{'Case 3 Plate Buckling Failure'})

subplot(2,3,4)
shear = plot(secA_locations(:,1),[-FOS_3A,-FOS_3A]*V, "r");
plot([100,100.0001], [-FOS_3A, -FOS_3B]*V, "r")
plot([100,285],[-FOS_3B,-FOS_3B]*V, "r-")
plot([285,285],[-FOS_3B,-FOS_3C]*V, "r-")
plot([285,915],[-FOS_3C,-FOS_3C]*V, "r-")
plot([915,915],[-FOS_3B,-FOS_3C]*V, "r-")
plot([915,1100],[-FOS_3B,-FOS_3B]*V, "r-")
plot([1100,1100.0001], [-FOS_3A, -FOS_3B]*V, "r")
plot(secA_locations(:,2),[-FOS_3A,-FOS_3A]*V, "r")

plot(secA_locations(:,1),[FOS_3A,FOS_3A]*V, "r")
plot([100,100.0001], [FOS_3A, FOS_3B]*V, "r")
plot([100,285],[FOS_3B,FOS_3B]*V, "r-")
plot([285,285],[FOS_3B,FOS_3C]*V, "r-")
plot([285,915],[FOS_3C,FOS_3C]*V, "r-")
plot([915,915],[FOS_3B,FOS_3C]*V, "r-")
plot([915,1100],[FOS_3B,FOS_3B]*V, "r-")
plot([1100,1100.0001], [FOS_3A, FOS_3B]*V, "r")
plot(secA_locations(:,2),[FOS_3A,FOS_3A]*V, "r")

legend([shear],{'Shear Failure'})

subplot(2,3,5)
glue = plot(secA_locations(:,1),[-FOS_4A,-FOS_4A]*V, "c");
plot([100,100.0001], [-FOS_4A, -FOS_4B]*V, "c")
plot([100,285],[-FOS_4B,-FOS_4B]*V, "c-")
plot([285,285],[-FOS_4B,-FOS_4C]*V, "c-")
plot([285,915],[-FOS_4C,-FOS_4C]*V, "c-")
plot([915,915],[-FOS_4B,-FOS_4C]*V, "c-")
plot([915,1100],[-FOS_4B,-FOS_4B]*V, "c-")
plot([1100,1100.0001], [-FOS_4A, -FOS_4B]*V, "c")
plot(secA_locations(:,2),[-FOS_4A,-FOS_4A]*V, "c")

plot(secA_locations(:,1),[FOS_4A,FOS_4A]*V, "c")
plot([100,100.0001], [FOS_4A, FOS_4B]*V, "c")
plot([100,285],[FOS_4B,FOS_4B]*V, "c-")
plot([285,285],[FOS_4B,FOS_4C]*V, "c-")
plot([285,915],[FOS_4C,FOS_4C]*V, "c-")
plot([915,915],[FOS_4B,FOS_4C]*V, "c-")
plot([915,1100],[FOS_4B,FOS_4B]*V, "c-")
plot([1100,1100.0001], [FOS_4A, FOS_4B]*V, "c")
plot(secA_locations(:,2),[FOS_4A, FOS_4A]*V, "c")

legend([glue],{'Glue Failure'})

subplot(2,3,6)
case4 = plot(secA_locations(:,1),[-FOS_8A,-FOS_8A]*V, "m");
```

```
    plot([100,100.0001], [-FOS_8A, -FOS_8B]*V, "m")
    plot([100,285],[-FOS_8B,-FOS_8B]*V, "m-")
    plot([285,285],[-FOS_8B,-FOS_8C]*V, "m-")
    plot([285,915],[-FOS_8C,-FOS_8C]*V, "m-")
    plot([915,915],[-FOS_8B,-FOS_8C]*V, "m-")
    plot([915,1100],[-FOS_8B,-FOS_8B]*V, "m-")
    plot([1100,1100.0001], [-FOS_8A, -FOS_8B]*V, "m")
    plot(secA_locations(:,2),[-FOS_8A,-FOS_8A]*V, "m")

    plot(secA_locations(:,1),[FOS_8A,FOS_8A]*V, "m")
    plot([100,100.0001], [FOS_8A, FOS_8B]*V, "m")
    plot([100,285],[FOS_8B,FOS_8B]*V, "m-")
    plot([285,285],[FOS_8B,FOS_8C]*V, "m-")
    plot([285,915],[FOS_8C,FOS_8C]*V, "m-")
    plot([915,915],[FOS_8B,FOS_8C]*V, "m-")
    plot([915,1100],[FOS_8B, FOS_8B]*V, "m-")
    plot([1100,1100.0001], [FOS_8A, FOS_8B]*V, "m")
    plot(secA_locations(:,2),[FOS_8A, FOS_8A]*V, "m")

    legend([case4],{'Case 4 Plate Buckling Failure'})



end
```